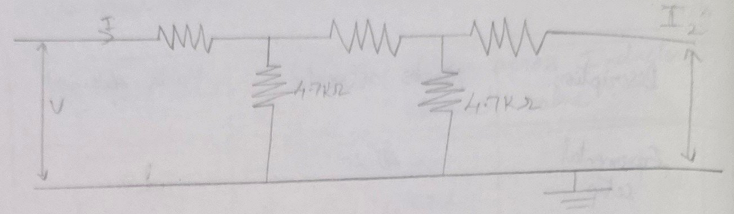
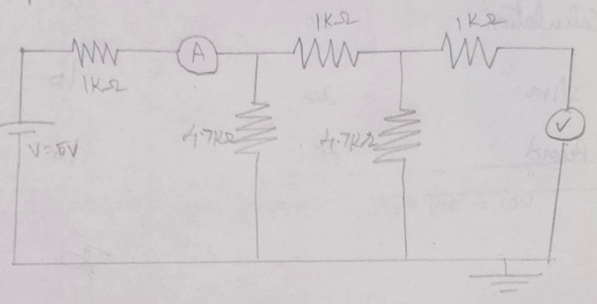
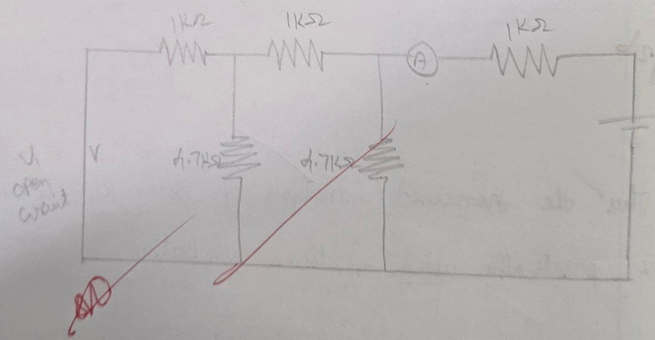


i) Output open circuit



ii) Input open circuit



Aim:

To

using m

Apparatu

S No	
1.	
2.	
3.	
4.	
5.	
6	

Procedure

* Connect
board
* First
O/P ter
* Secondly
O/P

Ex.No: 9
7/3/26

EXPERIMENTAL DETERMINATION OF Z PARAMETERS

Aim:

To calculate and verify Z parameters of a two part network using multisim simulator

Apparatus required:

S No	Component	Specification	Qty
1.	Powersupply	–	1
2.	Multimeter	–	1
3.	Resistor	10KΩ,	
4.	Bread board	—	1
5.	wires	–	–
6.	PC with multisim simulator	–	–

Procedure:

* Connect the circuit as shown in figure, switch on the experimental board
* First open the O/P terminal supply or to I/P terminal. Measure the O/P terminal voltage and I/P current
* Secondly open I/P terminal as supply to O/P terminal Measure I/P voltage O/P current

Observation:

S.No	Output open circuit ($I_2=0$)			$Z_{21} = V_2/I_1$ (Ω)	$Z_{11} = V_1/I_1$ (Ω)
	V_1	V_2	I_1		
1.	5	2.6	2.1	1.2[illegible]3 KΩ	2.38 KΩ
2.	5	2.55	2.03	1.22 KΩ	2.40 KΩ
3.	5	2.63	2.12	1.22 KΩ	2.33 KΩ

S.No	Input open circuit ($I_1=0$)			$Z_{12} = V_1/I_2$ (Ω)	$Z_{22} = V_2/I_2$ (Ω)
	V_2	V_1	I_2		
1.	5	2.6	2.1	1.29	4
2.	5	2.57	2.09	1.22	2.39
3.	5	2.62	2.12	1.24	2.38

verified

* calculate the values of z parameter using eqn ①&②
* Switch off the supply after taking the reading

Marks split up:

Sl No	Description	Marks allocated	Marks obtained
1.	Experimental setup	20	20
2.	Observation	20	20
3.	Calculation	20	10
4.	Viva	20	10
5.	Record	20	10
		Total	70

70

70
23/3/25

Result:

The z parameters of 2 part network has been calculated & verified using multisim simulator